

Experiment 10: Insertion Sort

Aim

To implement the **Insertion Sort** algorithm using Python to sort a list of elements in ascending order.

Algorithm

Step 1: Start.

Step 2: Read the number of elements n.

Step 3: Read the array elements.

Step 4: Assume the first element is already sorted.

Step 5: Compare the current element with the previous elements.

Step 6: Shift the larger elements one position to the right.

Step 7: Insert the current element in its correct position.

Step 8: Repeat Steps 5–7 until all elements are sorted.

Step 9: Display the sorted array.

Step 10: Stop.

Source Code (Python)

```
main.py    Share  Run
```

```
1 # Insertion Sort
2
3 n = int(input("Enter number of elements: "))
4
5 arr = []
6
7 print("Enter the elements:")
8 for i in range(n):
9     arr.append(int(input()))
10
11 for i in range(1, n):
12     key = arr[i]
13     j = i - 1
14
15     while j >= 0 and arr[j] > key:
16         arr[j + 1] = arr[j]
17         j -= 1
18
19     arr[j + 1] = key
20
21 print("Sorted array:", arr)
```

Output

Output

```
Enter number of elements: 6
Enter the elements:
45
10
20
80
50
30
Sorted array: [10, 20, 30, 45, 50, 80]

=== Code Execution Successful ===
```

Time Complexity

- Best Case: $O(n)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$

Space Complexity

- $O(1)$

Result

The Insertion Sort algorithm was implemented successfully in Python. The program sorts the given list of elements in ascending order by inserting each element into its correct position. The output was obtained correctly.